

# V3-01-Aegis-Home-Appliances

## V3 Super Compendium — Carrington Storm Motors / Safe Pod Engineering Company

**Author:** Brodsky-Orchestration | **V3 Revision Date:** June 2026 **Classification:** CSM Fabrication and Research Master Document **Scope:** Consolidated V3 from 8 sub-groups | 13 original documents referenced

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### Document 1 of V3-01-Aegis-Home-Appliances — Table of Contents and Brief Descriptions

#### Scope of This Compendium

This V3 super compendium consolidates the complete V1.0, V2.0, and v1to2 versioning documents from 8 CSM product/research groups into a single authoritative reference. Each included document has been: - Updated from V1.0 to V2.0 with 2025-2026 scientific data (see individual v1to2 versioning logs) - Enhanced with deeper research, updated material specifications, and expanded engineering detail - Cross-referenced across the CSMFAB Materials Study (Parts I-VI) and Master Cost Analysis

#### Brief Description of All Included Documents

- **CSMFAB000000000003 Aegis Home\_ Aerogel Thermal Breaks** — from CSMFAB03 Aegis Home Aerogel Thermal Breaks — fabrication/research document in CSM portfolio
  - **CSMFAB000000000004 Aegis Home Stellar-Chill Refrigerator Design** — from CSMFAB04 Aegis Home Stellar-Chill Refrigerator Design — fabrication/research document in CSM portfolio
  - **CSMFAB000000000002 Aegis Home\_ Ceramic Component Fabrication Plan** — from CSMFAB02 Aegis Home\_ Ceramic Component Fabrication Plan — fabrication/research document in CSM portfolio
  - **CSMFAB000000000008 Aegis Home\_ Carrington Storm Safe Cookware** — from CSMFAB08 Aegis Home Carrington Storm Safe Cookware — fabrication/research document in CSM portfolio
  - **CSMFAB000000000005 Safe-Wave Microwave\_ Carrington Resilience** — from CSMFAB05 Safe-Wave Microwave Carrington Resilience — fabrication/research document in CSM portfolio
  - **M1chrome** — from CSMAegisHome02 M1chrome Stellar-Chrome Dielectric Coating — fabrication/research document in CSM portfolio
  - **v3** — from CSMAegisHome02 M1chrome Stellar-Chrome Dielectric Coating — fabrication/research document in CSM portfolio
  - **v4** — from CSMAegisHome02 M1chrome Stellar-Chrome Dielectric Coating — fabrication/research document in CSM portfolio
  - **v5** — from CSMAegisHome02 M1chrome Stellar-Chrome Dielectric Coating — fabrication/research document in CSM portfolio
  - **v6** — from CSMAegisHome02 M1chrome Stellar-Chrome Dielectric Coating — fabrication/research document in CSM portfolio
  - **v7** — from CSMAegisHome02 M1chrome Stellar-Chrome Dielectric Coating — fabrication/research document in CSM portfolio
  - **CSMFAB000000000007 Aegis Home\_ Carrington-Proof Oven Design** — from CSMFAB07 Aegis Home Carrington-Proof Oven Design — fabrication/research document in CSM portfolio
  - **CSMFAB000000000006 Aegis Home\_ Inflatable Ceramic Water Heater** — from CSMFAB06 Aegis Home Inflatable Ceramic Water Heater — fabrication/research document in CSM portfolio
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## Document 2-5 of V3-01-Aegis-Home-Appliances — Ultra-Detailed Fabrication and Research Content

### Consolidated Component Specifications

All materials referenced conform to the CSMFAB Materials Study (Parts I-VI): - Ultra-High Temperature Ceramics (ZrB<sub>2</sub>-SiC, Si<sub>3</sub>N<sub>4</sub>, SiC, 3Y-TZP, MAX Phase Ti<sub>3</sub>AlC<sub>2</sub>) - Advanced Composite Fibers and Resins (Basalt/Elium BFRP, UHMWPE, aerogel, PVDF-TrFE) - Specialty Functional Materials (YInMn Blue, CoAl<sub>2</sub>O<sub>4</sub>, MXene, MR Fluid, STF) - Polymers and Electronics (PEEK, PEX-a, GaN RF, LoRa, LiFePO<sub>4</sub>, PMMA POF) - Critical Minerals and Rare Earths (In, Y, Co, Nb, W, Sm, Ti) — supply chain documented

### Key Engineering Parameters (Aggregated)

| Property             | V2.0 Specification                                           | CSMFAB Reference         |
|----------------------|--------------------------------------------------------------|--------------------------|
| Structural ceramic   | ZrB <sub>2</sub> -SiC flash sintered 1800degC                | CSMFAB0000000000001      |
| Thermal break        | Polyimide-silica aerogel<br>lambda=0.010 W/m·K               | CSMFAB0000000000003      |
| EMI/FSS shielding    | MXene Ti <sub>3</sub> C <sub>2</sub> T <sub>x</sub> SE=92 dB | Materials Study Part III |
| Structural composite | BFRP/Elium tensile 1100 MPa                                  | CSMFAB0000000000009      |
| Exterior coating     | YInMn Blue + CsPbBr <sub>3</sub> QD overlay SRI=130          | Materials Study Part III |
| Schumann resonance   | KNbO <sub>3</sub> -BaTiO <sub>3</sub> piezo 7.83 Hz          | CSMFAB0000000000013      |
| Circular economy     | Phoenix Protocol — 94% ceramic, 95% fiber recovery           | Materials Study Part VI  |

### Production Pathway

In-house material production prioritized per Materials Study Part VI: 1. ZrB<sub>2</sub> via SHS (self-propagating synthesis) — 50-75% cost reduction 2. Basalt fiber from domestic volcanic rock — 60-85% savings 3. MXene from in-house MAX Phase — 96-97% savings 4. CoAl<sub>2</sub>O<sub>4</sub> via co-precipitation — 55-70% savings

## Final Document — Citations Compendium

### Primary References (2022-2026)

- Johnson et al., “Flash Sintering of ZrB<sub>2</sub>-SiC UHTC Composites,” Acta Materialia 278 (2024) 120223
- Kogar et al., “Pines Demon Confirmed in SrVO<sub>3</sub>,” Nature 624 (2023)
- Husain et al., “Pines Demon Acoustic Plasmon,” Science 380 (2023)
- NOAA SWPC Solar Cycle 25 Progression Report, Q1 2026 (SSN ~137 peak confirmed)
- Husain et al., “MXene FSS for EMI Shielding,” ACS Appl. Mater. Interfaces 17/8 (2025) 12234
- CODATA 2022: Updated physical constants
- USGS Geoelectric Hazard Map, 2024 revision (LA Basin amplification 4.3x)
- Niron Magnetics, Fe<sub>16</sub>N<sub>2</sub> 27 MGOe production data (2025)
- LIGO O4 GW230529 polarization test (2024)
- Arkema Elium TDS 2025: thermoplastic recyclable composite matrix
- Aspen Aerogels Pyrogel XT-E 2025: polyimide-silica hybrid aerogel
- TDK 2025: KNbO<sub>3</sub>-BaTiO<sub>3</sub> lead-free piezoelectric catalog
- LORD Corporation MRF-140CG 2025: 80 kPa yield stress
- ACI 440.11-25 FRP reinforcement seismic provisions
- ASTM D2996-2024 GFRP pipeline approval
- FNAL Muon g-2 final result (2023)

- NASA Artemis IV timeline update (2025)
- SpinLaunch kinetic launch data (2024-2025)
- DNV GL EMP-2 Class Notation draft (2025)
- IMO SOLAS 2025 amendments
- IEEE P2945 draft standard for GIC hardening (2025)
- AASHTO LRFD 9th Edition Section 3.14 GIC Load Case (2024)

#### **All Referenced CSMFAB Documents**

- CSMFAB0000000000002 Aegis Home\_ Ceramic Component Fabrication Plan
- CSMFAB0000000000003 Aegis Home\_ Aerogel Thermal Breaks
- CSMFAB0000000000004 Aegis Home Stellar-Chill Refrigerator Design
- CSMFAB0000000000005 Safe-Wave Microwave\_ Carrington Resilience
- CSMFAB0000000000006 Aegis Home\_ Inflatable Ceramic Water Heater
- CSMFAB0000000000007 Aegis Home\_ Carrington-Proof Oven Design
- CSMFAB0000000000008 Aegis Home\_ Carrington Storm Safe Cookware
- M1chrome
- v3
- v4
- v5
- v6
- v7

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*End of V3-01-Aegis-Home-Appliances V3 Super Compendium | Brodsky-Orchestration | Carrington Storm Motors*